

Newman-Penrose Formalism Calculations

Stephani et al., Equation (36.26)

Metric

$$ds^2 = -e^{(2\nu)} \left(\frac{\partial T}{\partial t} \right)^2 dt \otimes dt - e^{(2\nu)} \frac{\partial T}{\partial t} \frac{\partial T}{\partial r} dt \otimes dr + r^2 d\theta \otimes d\theta + r^2 \sin^2(\theta) d\phi \otimes d\phi - e^{(2\nu)} \frac{\partial T}{\partial t} \frac{\partial T}{\partial r} dr \otimes dt + \left(-e^{(2\nu)} \left(\frac{\partial T}{\partial r} \right)^2 + e^{(2\lambda)} \right) dr \otimes dr$$

Coordinates

$$(x^\mu) = (t, \theta, \phi, r)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} e^\nu \frac{\partial}{\partial t} T(t, r) \right) dt + \left(\frac{1}{2} \sqrt{2} e^\nu \frac{\partial}{\partial r} T(t, r) - \frac{1}{2} \sqrt{2} e^\lambda \right) dr$$

$$n = \left(\frac{1}{2} \sqrt{2} e^\nu \frac{\partial}{\partial t} T(t, r) \right) dt + \left(\frac{1}{2} \sqrt{2} e^\nu \frac{\partial}{\partial r} T(t, r) + \frac{1}{2} \sqrt{2} e^\lambda \right) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r \right) d\theta + \left(\frac{1}{2} i \sqrt{2} r \sin(\theta) \right) d\phi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r \right) d\theta + \left(-\frac{1}{2} i \sqrt{2} r \sin(\theta) \right) d\phi$$

Contravariant components

$$l = \left(\frac{\sqrt{2} e^{(-\lambda)} \frac{\partial}{\partial r} T(t, r)}{2 \frac{\partial}{\partial t} T(t, r)} - \frac{\sqrt{2} e^{(-\nu)}}{2 \frac{\partial}{\partial t} T(t, r)} \right) dt + \left(-\frac{1}{2} \sqrt{2} e^{(-\lambda)} \right) dr$$

$$n = \left(-\frac{\sqrt{2}e^{(-\lambda)} \frac{\partial}{\partial r} T(t, r)}{2 \frac{\partial}{\partial t} T(t, r)} - \frac{\sqrt{2}e^{(-\nu)}}{2 \frac{\partial}{\partial t} T(t, r)} \right) dt + \left(\frac{1}{2} \sqrt{2}e^{(-\lambda)} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2r} \right) d\theta + \left(\frac{i\sqrt{2}}{2r \sin(\theta)} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2r} \right) d\theta + \left(-\frac{i\sqrt{2}}{2r \sin(\theta)} \right) d\phi$$

Directional Derivatives

$$D = \frac{\sqrt{2}e^{(-\lambda)} \frac{\partial}{\partial r} T(t, r)}{2 \frac{\partial}{\partial t} T(t, r)} - \frac{\sqrt{2}e^{(-\nu)}}{2 \frac{\partial}{\partial t} T(t, r)} \partial_t - \frac{1}{2} \sqrt{2}e^{(-\lambda)} \partial_r$$

$$\Delta = -\frac{\sqrt{2}e^{(-\lambda)} \frac{\partial}{\partial r} T(t, r)}{2 \frac{\partial}{\partial t} T(t, r)} - \frac{\sqrt{2}e^{(-\nu)}}{2 \frac{\partial}{\partial t} T(t, r)} \partial_t + \frac{1}{2} \sqrt{2}e^{(-\lambda)} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2r} \partial_\theta + \frac{i\sqrt{2}}{2r \sin(\theta)} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2r} \partial_\theta - \frac{i\sqrt{2}}{2r \sin(\theta)} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}e^{(-\lambda)}}{2r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}e^{(-\lambda)}}{2\,r}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cos\left(\theta\right)}{4\,r\sin\left(\theta\right)}$$

$$\beta = \frac{\sqrt{2}\cos\left(\theta\right)}{4\,r\sin\left(\theta\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{e^{(-2\lambda)}}{4r^2} + \frac{1}{4r^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = -\frac{e^{(-2\lambda)}}{12r^2} + \frac{1}{12r^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{e^{(-2\lambda)}}{6r^2} - \frac{1}{6r^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "*D*"